Week assessment 7 :Student Management System Database

```
CREATE TABLE Students (
    student_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE,
    phone VARCHAR(15),
    dob DATE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
INSERT INTO Students (name, email, phone, dob) VALUES
('Asha', 'asha@gmail.com', '9876543210', '2002-05-10'),
('Rahul', 'rahul@gmail.com', '9876543211', '2001-03-15'),
('Sneha', 'sneha@gmail.com', '9876543212', '2002-07-20'),
('Kiran', 'kiran@gmail.com', '9876543213', '2001-09-11'),
('Meena', 'meena@gmail.com', '9876543214', '2003-01-25'),
('Arjun', 'arjun@gmail.com', '9876543215', '2002-02-18'),
('Divya', 'divya@gmail.com', '9876543216', '2001-06-30'),
('Vikram', 'vikram@gmail.com', '9876543217', '2002-12-05'),
('Pooja', 'pooja@gmail.com', '9876543218', '2003-08-19'),
('Ravi', 'ravi@gmail.com', '9876543219', '2001-11-22');
```

	student_id	name	email	phone	dob	created_at
Edit Copy Delete	1	Asha	asha@gmail.com	9876543210	2002-05-10	2026-03-30 06:57:10
Edit Copy Delete	2	Rahul	rahul@gmail.com	9876543211	2001-03-15	2026-03-30 06:57:10
Edit Copy Delete	3	Sneha	sneha@gmail.com	9876543212	2002-07-20	2026-03-30 06:57:10
Edit Copy Delete	4	Kiran	kiran@gmail.com	9876543213	2001-09-11	2026-03-30 06:57:10
Edit Copy Delete	5	Meena	meena@gmail.com	9876543214	2003-01-25	2026-03-30 06:57:10
Edit Copy Delete	6	Arjun	arjun@gmail.com	9876543215	2002-02-18	2026-03-30 06:57:10
Edit Copy Delete	7	Divya	divya@gmail.com	9876543216	2001-06-30	2026-03-30 06:57:10
Edit Copy Delete	8	Vikram	vikram@gmail.com	9876543217	2002-12-05	2026-03-30 06:57:10
Edit Copy Delete	9	Pooja	pooja@gmail.com	9876543218	2003-08-19	2026-03-30 06:57:10
Edit Copy Delete	10	Ravi	ravi@gmail.com	9876543219	2001-11-22	2026-03-30 06:57:10

Courses

```
CREATE TABLE Courses (
    course_id INT PRIMARY KEY AUTO_INCREMENT,
    course_name VARCHAR(100) NOT NULL,
    credits INT DEFAULT 3
);
INSERT INTO Courses (course_name, credits) VALUES
('Maths', 4),
('Physics', 3),
('Chemistry', 3),
('Computer Science', 5),
('English', 2);
```

←T→				course_id	course_name	credits
☐	Edit	Copy	Delete	1	Maths	4
☐	Edit	Copy	Delete	2	Physics	3
☐	Edit	Copy	Delete	3	Chemistry	3
☐	Edit	Copy	Delete	4	Computer Science	5
☐	Edit	Copy	Delete	5	English	2

Enrollments

```
CREATE TABLE Enrollments (
    enrollment_id INT PRIMARY KEY AUTO_INCREMENT,
    student_id INT,
    course_id INT,
    enrollment_date DATE,
    FOREIGN KEY (student_id) REFERENCES Students(student_id),
    FOREIGN KEY (course_id) REFERENCES Courses(course_id)
);
INSERT INTO Enrollments (student_id, course_id, enrollment_date) VALUES
```

(1,1,'2024-01-10'), (2,2,'2024-01-11'),

(3,3,'2024-01-12'), (4,4,'2024-01-13'),

(5,5,'2024-01-14'), (6,1,'2024-01-15'),

(7,2,'2024-01-16'), (8,3,'2024-01-17'),

(9,4,'2024-01-18'), (10,5,'2024-01-19'),

(1,2,'2024-01-20'), (2,3,'2024-01-21'),

(3,4,'2024-01-22'), (4,5,'2024-01-23'),

(5,1,'2024-01-24');

←T→	enrollment_id	student_id	course_id	enrollment_date
Edit Copy Delete	1	1	1	2024-01-10
Edit Copy Delete	2	2	2	2024-01-11
Edit Copy Delete	3	3	3	2024-01-12
Edit Copy Delete	4	4	4	2024-01-13
Edit Copy Delete	5	5	5	2024-01-14
Edit Copy Delete	6	6	1	2024-01-15
Edit Copy Delete	7	7	2	2024-01-16
Edit Copy Delete	8	8	3	2024-01-17
Edit Copy Delete	9	9	4	2024-01-18
Edit Copy Delete	10	10	5	2024-01-19
Edit Copy Delete	11	1	2	2024-01-20
Edit Copy Delete	12	2	3	2024-01-21
Edit Copy Delete	13	3	4	2024-01-22
Edit Copy Delete	14	4	5	2024-01-23
Edit Copy Delete	15	5	1	2024-01-24

Marks

```
CREATE TABLE Marks (
    mark_id INT PRIMARY KEY AUTO_INCREMENT,
    student_id INT,
    course_id INT,
    marks INT,
    grade VARCHAR(2),
```

```
    FOREIGN KEY (student_id) REFERENCES Students(student_id),
    FOREIGN KEY (course_id) REFERENCES Courses(course_id)
);
INSERT INTO Marks (student_id, course_id, marks, grade) VALUES
(1,1,85,'A'), (2,2,78,'B'), (3,3,90,'A'),
(4,4,67,'C'), (5,5,88,'A'), (6,1,76,'B'),
(7,2,82,'A'), (8,3,59,'D'), (9,4,91,'A'),
(10,5,73,'B'), (1,2,84,'A'), (2,3,69,'C'),
(3,4,88,'A'), (4,5,92,'A'), (5,1,75,'B'),
(6,2,81,'A'), (7,3,66,'C'), (8,4,79,'B'),
(9,5,85,'A'), (10,1,70,'B');
```

				mark_id	student_id	course_id	marks	grade
☐	Edit	Copy	Delete	1	1	1	85	A
☐	Edit	Copy	Delete	3	3	3	90	A
☐	Edit	Copy	Delete	4	4	4	67	C
☐	Edit	Copy	Delete	5	5	5	88	A
☐	Edit	Copy	Delete	6	6	1	76	B
☐	Edit	Copy	Delete	7	7	2	82	A
☐	Edit	Copy	Delete	8	8	3	59	D
☐	Edit	Copy	Delete	9	9	4	91	A
☐	Edit	Copy	Delete	10	10	5	73	B
☐	Edit	Copy	Delete	11	1	2	84	A
☐	Edit	Copy	Delete	12	2	3	69	C
☐	Edit	Copy	Delete	13	3	4	88	A
☐	Edit	Copy	Delete	14	4	5	92	A
☐	Edit	Copy	Delete	15	5	1	75	B
☐	Edit	Copy	Delete	16	6	2	81	A
☐	Edit	Copy	Delete	17	7	3	66	C
☐	Edit	Copy	Delete	18	8	4	79	B
☐	Edit	Copy	Delete	19	9	5	85	A
☐	Edit	Copy	Delete	20	10	1	70	B

Views

```
CREATE VIEW student_marks_view AS
SELECT s.name, m.marks
```

FROM Students s

JOIN Marks m ON s.student_id = m.student_id;

←T→				name	marks
☐	Edit	Copy	Delete	Asha	85
☐	Edit	Copy	Delete	Asha	84
☐	Edit	Copy	Delete	Rahul	69
☐	Edit	Copy	Delete	Sneha	90
☐	Edit	Copy	Delete	Sneha	88
☐	Edit	Copy	Delete	Kiran	67
☐	Edit	Copy	Delete	Kiran	92
☐	Edit	Copy	Delete	Meena	88
☐	Edit	Copy	Delete	Meena	75
☐	Edit	Copy	Delete	Arjun	76
☐	Edit	Copy	Delete	Arjun	81
☐	Edit	Copy	Delete	Divya	82
☐	Edit	Copy	Delete	Divya	66
☐	Edit	Copy	Delete	Vikram	59
☐	Edit	Copy	Delete	Vikram	79
☐	Edit	Copy	Delete	Pooja	91
☐	Edit	Copy	Delete	Pooja	85
☐	Edit	Copy	Delete	Ravi	73
☐	Edit	Copy	Delete	Ravi	70